

Exactness: DE looks like $\Psi_x dx + \Psi_y dy = 0$

for $\Psi(y, x) \in C^2$ (twice diff'ble) we have $\Psi_{xy} = \Psi_{yx}$

this is what we use to check if the DE is exact or not.

For $F(x) = \Psi(y(x), x)$

$$\frac{dF}{dx} = \frac{\partial \Psi}{\partial y} \frac{dy}{dx} + \frac{\partial \Psi}{\partial x} \quad (\text{total derivative})$$

$F'(x) = 0$, then $F(x) = C$ is a solution for $C \in \mathbb{R}$.

$$\Rightarrow \Psi(y(x), x) = C$$

In each of Problems 15 and 16, find the value of b for which the given equation is exact, and then solve it using that value of b .

2.6. 15. $(xy^2 + bx^2y) + (x+y)x^2y' = 0$

Does there exist a $\Psi(x, y)$ twice diff'ble, so that

$$\underbrace{(xy^2 + bx^2y)}_{\Psi_x} + \underbrace{(x+y)x^2y'}_{\Psi_y} = \Psi_x + \Psi_y \cdot y'$$

$$(xy^2 + bx^2y)_y \stackrel{?}{=} [(x+y)x^2]_x$$

$$2xy + bx^2$$

$$1 \cdot x^2 + (x+y)2x = \underbrace{x^2 + 2x^2}_{b=3} + 2xy$$

$$\Psi_x = xy^2 + 3x^2y, \quad \Psi_y = x^3 + yx^2$$

$$\Psi(y, x) = \frac{x^2}{2}y^2 + x^3y + f(y)$$

$$\Psi_y = x^2y + x^3 + f'(y)$$

$$f'(y) = 0 \quad f(y) = \text{const.}$$

$$\psi(x, y) = \frac{x^2 y^2}{2} + x^3 y + C$$

Soln: $\int (xy^2 + 6x^2y) + (x+y)x^2y' dx = \int 0 dx$

$$\frac{x^2 y^2}{2} + x^3 y + C_1 = C_2 \rightarrow \text{implicit s.ln.}$$

2.6. 18. Show that any separable equation

$$M(x) + N(y)y' = 0$$

exercise

is also exact.

$\mu(x, y)$ "MAKING A DE. EXACT"

$$M + N y' = 0 \quad \cdot \mu(x, y)$$

$$\underbrace{\mu M}_{\psi_x} + \underbrace{\mu N}_{\psi_y} y' = 0$$

$$\text{exact} \Leftrightarrow (\mu M)_y = (\mu N)_x$$

$$\mu_y M + \mu M_y = \mu_x N + \mu N_x \quad *$$

If we assume $\mu = \mu(y) \rightarrow \begin{cases} \mu_y = \frac{d\mu}{dy} = \mu' \\ \mu_x = 0 \end{cases}$

$\textcircled{*}$ becomes: $\mu' M + \mu M_y = 0 + \mu N_x$

do not depend on x $\left\{ \begin{array}{l} \text{fnc of } y \leftarrow \mu' \\ \text{fnc of } y \leftarrow \mu \end{array} \right. \frac{\mu'}{\mu} = \frac{N_x - M_y}{M} = \text{fnc of } y$

If $\frac{N_x - M_y}{M}$ depends only on y , $\ln|\mu| = \int \frac{N_x - M_y}{\mu} dy$

2.6 24. Show that if $(N_x - M_y)/(xM - yN) = R$, where R depends on the quantity xy only, then the differential equation

$$M + Ny' = 0$$

has an integrating factor of the form $\mu(xy)$. Find a general formula for this integrating factor.

$$\text{exact} \Leftrightarrow (\mu M)_y = (\mu N)_x$$

$$\mu_y M + \mu M_y = \mu_x N + \mu N_x \quad *$$

$$\underline{\mu(xy)}: \text{ Call } xy = v$$

$$\mu_x = \frac{\partial \mu(v)}{\partial x} = \frac{d\mu}{dv} \cdot \frac{\partial v}{\partial x} \quad \begin{matrix} \text{"}\mu\text{"} \\ \text{"}v\text{"} \end{matrix}$$

$$\mu_y = \frac{\partial \mu(v)}{\partial y} = \frac{d\mu}{dv} \cdot \frac{\partial v}{\partial y} \quad \begin{matrix} \text{"}\mu\text{"} \\ \text{"}v\text{"} \end{matrix}$$

* becomes:

$$\underbrace{\mu'_x}_{=\mu_y} M + \mu M_y = \underbrace{\mu'_y}_{=\mu_x} N + \mu N_x$$

$$\mu'(xM - yN) = \mu(N_x - M_y)$$

$$\text{fnc of } v \leftarrow \frac{\mu'}{\mu} = \frac{N_x - M_y}{xM - yN} \quad \left. \vphantom{\frac{\mu'}{\mu}} \right\} = \text{fnc of } v$$

$$v = xy$$

$$\text{sth like} \quad \frac{(xy)^2 + 3xy}{4e^{xy}}$$

Assume $\frac{N_x - M_y}{xM - yN}$ is a fnc of $v = xy$

$$\ln |\mu| = \int \frac{N_x - M_y}{xM - yN} dv \quad \Rightarrow \quad \mu = e^{\int \frac{N_x - M_y}{xM - yN} dv}$$

2.6. In each of Problems 25 through 31, find an integrating factor and solve the given equation.

25. $(3x^2y + 2xy + y^3) + (x^2 + y^2)y' = 0$

28. $y + (2xy - e^{-2y})y' = 0 \rightarrow \text{exercise}$

$$\textcircled{25} \quad (3x^2y + 2xy + y^3)_y \stackrel{?}{=} (x^2 + y^2)_x$$

$3x^2 + 2x \qquad \qquad \qquad 2x \qquad \Rightarrow \text{not exact.}$

Let $\mu = \mu(x)$

$$[(3x^2y + 2xy + y^3)\mu(x)]_y = [(x^2 + y^2)\mu(x)]_x$$

$$(3x^2 + 2x + 3y^2)\mu(x) = (2x)\mu(x) + (x^2 + y^2)\mu'(x)$$

$$(3x^2 + \cancel{2x} - \cancel{2x} + 3y^2)\mu(x) = (x^2 + y^2)\mu'(x)$$

$$3 = \frac{3x^2 + 3y^2}{x^2 + y^2} = \frac{\mu'}{\mu} \quad \Rightarrow \text{Then } \mu \text{ is indeed } \mu = \mu(x)$$

$$\int 3 dx = \int \frac{\mu'}{\mu} dx$$

$$3x + C = \ln |\mu|$$

$$e^{3x} \cdot C = e^{|\mu|} \quad \Rightarrow \mu = e^{3x}$$

$$\underbrace{e^{3x}(3x^2y + 2xy + y^3)}_{\psi_x} + \underbrace{e^{3x}(x^2 + y^2)}_{\psi_y} y' = 0$$

↙ ↘

$$\psi(x, y) = e^{3x} \left(x^2y + \frac{y^3}{3} \right) + f(x)$$

$$\psi_x = 3e^{3x} \left(x^2 y + \frac{y^3}{3} \right) + e^{3x} (2xy) + f'(x)$$

$$\Rightarrow f'(x) = 0 \Rightarrow f(x) = c$$

$$\psi(x, y) = e^{3x} \left(x^2 y + \frac{y^3}{3} \right) + c$$

Soln. $\left(\psi = \text{const} \right)$ $e^{3x} \left(x^2 y + \frac{y^3}{3} \right) = \text{const.}$

(28) Hint: $\mu(y) = e^{2y} \cdot y^{-1}$ Soln: $x e^{2y} - \ln y = c$

Misc.
Prob's

33. Riccati Equations. The equation

$$\frac{dy}{dt} = q_1(t) + q_2(t)y + q_3(t)y^2$$

is known as a Riccati²³ equation. Suppose that some particular solution y_1 of this equation is known. A more general solution containing one arbitrary constant can be obtained through the substitution

$$y = y_1(t) + \frac{1}{v(t)},$$

Show that $v(t)$ satisfies the first order linear equation

$$\frac{dv}{dt} = -(q_2 + 2q_3 y_1)v - q_3.$$

Note that $v(t)$ will contain a single arbitrary constant.

y_1 is a soln:

$$y_1' = q_1 + q_2 y_1 + q_3 y_1^2$$

$$y = y_1(t) + \frac{1}{v(t)} \xrightarrow{\text{diff.}} y' = y_1' - \frac{v'}{v^2}$$

$$\begin{aligned} \underline{y_1'} - \frac{v'}{v^2} &= q_1 + q_2 \left(y_1 + \frac{1}{v} \right) + q_3 \left(y_1 + \frac{1}{v} \right)^2 \\ &= \underline{q_1} + \underline{q_2 y_1} + \frac{q_2}{v} + \underline{q_3 y_1^2} + 2q_3 \frac{y_1}{v} + \frac{q_3}{v^2} \end{aligned}$$

$$-\frac{v'}{v^2} = \frac{q_2}{v} + 2q_3 \frac{y_1}{v} + \frac{q_3}{v^2} \quad (\cdot (-v^2))$$

$$v' = -(q_2 + 2q_3 y_1)v - q_3$$

Misc.
Prob.

34. Using the method of Problem 33 and the given particular solution, solve each of the following Riccati equations:

$$(a) \quad y' = \underbrace{1}_{q_1} + \underbrace{t^2}_{q_2} - \underbrace{2ty}_{q_3} + \underbrace{y^2}_{q_4}; \quad y_1(t) = t$$

$$y = y_1 + \frac{1}{v}$$

$$v' = -(\underbrace{-2t + 2t}_{=0})v - 1 \Rightarrow v' = -1$$

$$\Rightarrow v = -t + C$$

$$\Rightarrow y = t + \frac{1}{C-t}$$

1st order DE:

- linear $\rightarrow \mu(x)$ integrating factor
- nonlinear:
 - separable $\rightarrow \int A(x)dx = \int B(y)dy$
 - exact $\rightarrow$ Find ψ s.t. $\psi_x dx + \psi_y dy = 0$
soln: $\psi = \text{constant}$.
 - not exact $\rightarrow$ Make the DE exact & solve.
 $\left\{ \begin{array}{l} \text{Is there } \mu(x,y) \text{ s.t.} \\ \mu \cdot (\text{DE}) \text{ is exact?} \end{array} \right\}$
- nothing works: Is it homogeneous?

$$y' = F(x,y) \stackrel{?}{=} F\left(\frac{y}{x}\right) \quad \text{if true} \quad = F\left(\frac{\lambda y}{\lambda x}\right)$$

check: $F(x,y) \stackrel{?}{=} F(\lambda x, \lambda y)$

if homogeneous, $v = \frac{y}{x} \Rightarrow \text{DE}(v)$ is separable. Solve.

In each of Problems 1 and 2, transform the given initial value problem into an equivalent problem with the initial point at the origin.

2.8. 1. $dy/dt = t^2 + y^2$, $y(1) = 2$

$t \mapsto s$
 $y \mapsto z$ s.t. $z(0) = 0$

$s+1=t$

If we let $s=t-1$ and define $z=y-2$. Then

$s=0 \Rightarrow t=1 \Rightarrow y=2 \Rightarrow z=0$.

Now we write the DE $z(s)$ satisfies:

$$\frac{dz(s)}{ds} = \frac{dz(t(s))}{dt} \frac{dt(s)}{ds} = \frac{d(y-2)}{dt} \cdot 1 = \frac{dy}{dt} = t^2 + y^2 = (s+1)^2 + (z+2)^2$$

So we have

$z' = \frac{dz}{ds} = (s+1)^2 + (z+2)^2$ with IV: $z(0)=0$.

In each of Problems 3 through 6, let $\phi_0(t) = 0$ and define $\{\phi_n(t)\}$ by the method of successive approximations

(a) Determine $\phi_n(t)$ for an arbitrary value of n .

~~(b) Plot $\phi_n(t)$ for $n=1, \dots, 4$. Observe whether the iterates appear to be converging.~~

(c) Express $\lim_{n \rightarrow \infty} \phi_n(t) = \phi(t)$ in terms of elementary functions; that is, solve the given initial value problem.

2.8. 3. $y' = 2(y+1)$, $y(0) = 0$

(a) We'll use $\phi_{n+1}(t) = \int_0^t 2(\phi_n(s)+1) ds$.

Start with $\phi_0(t)=0$, $\phi_1(t) = \int_0^t 2(\phi_0(s)+1) ds = \int_0^t 2 ds = [2s]_{s=0}^t = 2t$

$\phi_2(t) = \int_0^t 2(\phi_1(s)+1) ds = \int_0^t 4s+2 ds = [2s^2+2s]_{s=0}^t = 2t^2 + 2t$ ϕ_1

$\phi_3(t) = \int_0^t 2(\phi_2(s)+1) ds = \int_0^t 4s^2+4s+2 ds = [\frac{4}{3}s^3+2s^2+2s]_{s=0}^t = \frac{4t^3}{3} + 2t^2 + 2t$ ϕ_2

Note that $2t^2 = \int_0^t 2 \cdot (2s) ds$ and $\frac{4t^3}{3} = \int_0^t 2 \cdot (2s^2) ds$

So for Φ_4 , we will add $\int_0^t 2 \cdot \frac{4s^3}{3} ds$ to Φ_3 , and so on:

$$\phi_1 = 2t$$

$$\phi_2 = 2 \cdot t^2 + \phi_1$$

$$\phi_3 = 2 \cdot 2 \cdot \frac{t^3}{3} + \phi_2$$

$$\phi_4 = 2 \cdot 2 \cdot 2 \cdot \frac{t^4}{3 \cdot 4} + \phi_3$$

$$\phi_5 = 2 \cdot 2 \cdot 2 \cdot 2 \cdot \frac{t^5}{3 \cdot 4 \cdot 5} + \phi_4 = 2^5 \cdot \frac{t^5}{2 \cdot 3 \cdot 4 \cdot 5} + \phi_4 = 2^5 \frac{t^5}{5!} + \phi_4$$

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$$\phi_k = 2^k \frac{t^k}{k!} + \phi_{k-1}$$

$$= 2^k \frac{t^k}{k!} + 2^{k-1} \frac{t^{k-1}}{(k-1)!} + 2^{k-2} \frac{t^{k-2}}{(k-2)!} + \dots + 2^2 \frac{t^2}{2} + 2 \cdot \frac{t}{1!}$$

$$= \sum_{i=1}^k \frac{2^i t^i}{i!}$$

You can use mathematical induction to prove it.

• Basis step: $\mathbb{I}_1$ ✓

- Inductive step: Assume $\phi_k = \sum_{i=1}^k \frac{2^i t^i}{i!}$, show $\phi_{k+1} = \sum_{i=1}^{k+1} \frac{2^i t^i}{i!}$

$$\phi_{k+1} = \int_0^t 2(\phi_k(s) + 1) ds = \int_0^t 2 \sum_{i=1}^k \frac{2^i s^i}{i!} + 2 ds = \int_0^t \sum_{i=1}^k \frac{2^{i+1} s^i}{i!} ds + 2t$$

$$= \sum_{i=1}^k \int_0^t \frac{2^{i+1} s^i}{i!} ds + 2t = \sum_{i=1}^k \frac{2^{i+1}}{i!} \cdot \left[\frac{s^{i+1}}{(i+1)} \right]_{s=0}^t + 2t = \sum_{i=1}^k \frac{2^{i+1} t^{i+1}}{(i+1)!} + 2t$$

$$(c) \lim_{n \rightarrow \infty} \sum_{i=1}^n \frac{(2t)^i}{i!} + 1 - 1 = \lim_{n \rightarrow \infty} \sum_{i=0}^n \frac{(2t)^i}{i!} - 1 = e^{2t} - 1$$

$$= \sum_{j=1}^{k+1} \frac{2^j \cdot t^j}{j!}$$

In each of Problems 1 through 8, find the general solution of the given differential equation.

3.1. 2. $y'' + 3y' + 2y = 0$

2nd order, linear, ODE

hom. eqn. w/ const. coef.

$$y'' - y = 0$$

$$y'' = y$$

$$e^t$$

$$e^{-t}$$

$$y = e^{rt}, \quad y' = r e^{rt}, \quad y'' = r^2 e^{rt}$$

$$y'' + ay' + by = 0$$

$$\Rightarrow \underbrace{(r^2 + ar + b)}_{\text{char. eqn.} = 0} \underbrace{e^{rt}}_{\neq 0} = 0$$

y_1 soln. $\Rightarrow c y_1$ soln.

$$(c y_1)' = c y_1' \quad (c y_1)'' = c y_1''$$

$$c_1 (y_1'' + a y_1' + b y_1) = 0$$

$$+ c_2 (y_2'' + a y_2' + b y_2) = 0$$

$$c_1 y_1 + c_2 y_2$$

char. eqn: $r^2 + 3r + 2 = 0 \Rightarrow r_1 = -2 \quad r_2 = -1$

$$(r+1)(r+2)$$

$$\downarrow \quad \downarrow$$

$$y_1 = c_1 e^{-2t}$$

$$y_2 = c_2 e^{-t}$$

general soln: $y = c_1 e^{-2t} + c_2 e^{-t}$

In each of Problems 9 through 16, find the solution of the given initial value problem. Sketch the graph of the solution and describe its behavior as t increases.

3.1. 10. $y'' + 4y' + 3y = 0, \quad y(0) = 2, \quad y'(0) = -1$

char. eqn: $r^2 + 4r + 3 = 0 \Rightarrow r_1 = -1 \quad r_2 = -3$

$$\downarrow$$

$$y_1 = c_1 e^{-t}$$

$$\downarrow$$

$$y_2 = c_2 e^{-3t}$$

general soln: $y = c_1 e^{-t} + c_2 e^{-3t}$

$$\rightarrow y' = -c_1 e^{-t} - 3c_2 e^{-3t}$$

IVS: $y(0) = 2 \Rightarrow c_1 + c_2 = 2$

$$y'(0) = -1 \Rightarrow -c_1 - 3c_2 = -1$$

$$-2c_2 = 1$$

$$c_2 = -1/2$$

$$c_1 = 5/2$$

soln to the IVP: $y = \frac{5}{2} e^{-t} - \frac{1}{2} e^{-3t}$

3.1.17. Find a differential equation whose general solution is $y = c_1 e^{2t} + c_2 e^{-3t}$.

char eqn: $(r-2)(r+3) = 0$

$\downarrow \quad \downarrow$
 $r_1 = 2 \quad r_2 = -3$

$r^2 + r - 6 = 0$

DE: $y'' + y' - 6y = 0$

3.1.21. Solve the initial value problem $y'' - y' - 2y = 0$, $y(0) = \alpha$, $y'(0) = 2$. Then find α so that the solution approaches zero as $t \rightarrow \infty$.

char eqn: $r^2 - r - 2 = 0$
 $(r-2)(r+1)$

$\Rightarrow r_1 = 2, r_2 = -1$
 $\downarrow \quad \downarrow$

$y_1 = c_1 e^{2t} \quad y_2 = c_2 e^{-t}$

general soln: $y = c_1 e^{2t} + c_2 e^{-t}$
 $y' = 2c_1 e^{2t} - c_2 e^{-t}$

IVs: $y(0) = \alpha \Rightarrow c_1 + c_2 = \alpha$
 $y'(0) = 2 \Rightarrow 2c_1 - c_2 = 2$

$$\left. \begin{array}{l} c_1 + c_2 = \alpha \\ 2c_1 - c_2 = 2 \end{array} \right\} \begin{array}{l} c_1 = \frac{\alpha+2}{3} \\ c_2 = \frac{2\alpha-2}{3} \end{array}$$

$3c_1 = \alpha + 2$

Soln: $y = \frac{\alpha+2}{3} e^{2t} + \frac{2\alpha-2}{3} e^{-t}$
 $\downarrow t \rightarrow \infty \quad \downarrow t \rightarrow \infty$
 $\alpha \quad 0$

$\frac{\alpha+2}{3} = 0 \quad \alpha = -2$